Mathematics practice booklet: Grade 8 expectations

Grade 8 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 8 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 8.EE.B.5, 8.EE.B.6, 8.EE.C.7, 8.EE.C.8, 8.F.A.1, 8.F.A.2, 8.F.A.3, 8.F.B.4

A worked example

Solve: 3(x - 4) = 2x + 5. Then check your answer.

1. Expand the bracket: 3x - 12 = 2x + 5.
2. Take 2x from both sides: x - 12 = 5.
3. Add 12 to both sides: x = 17.
4. Check: 3 x (17 - 4) = 39 and 2 × 17 + 5 = 39. Both sides match.

x = 17

Questions

1. Solve: 3x + 7 = 28

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2. Solve: 5(x - 2) = 3x + 6

   A. x = 8

   B. x = 4

   C. x = 16

   D. x = -2

3. Hind thinks of a number, halves it and then adds 3. The result is 11. What is her number?

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4. What can you say about the equation 2x + 3 = 2x + 5?

   A. It has infinitely many solutions

   B. x = 0

   C. x = 1

   D. It has no solution

5. What is the slope of the line through (1, 2) and (4, 11)?

   A. 9

   B. 1/3

   C. 3

   D. -3

6. Write the equation of the line with slope 2 that crosses the y-axis at (0, -3).

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7. A taxi in Dubai charges AED 12 to start and AED 3 for each km. Write the cost y for x km, then find the cost of 10 km.

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8. Which pair (x, y) solves both y = 2x + 1 and y = x + 4?

   A. (7, 3)

   B. (1, 5)

   C. (3, 4)

   D. (3, 7)

9. Maryam and Noura have 10 stickers together. Maryam has 4 more than Noura. Use x + y = 10 and x - y = 4 to find how many each has.

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10. Is {(1, 3), (2, 5), (2, 7)} a function?

    A. Yes, because every input is used

    B. No, because the input 2 has two outputs, 5 and 7

    C. Yes, because it has three pairs

    D. No, because the outputs are all different

11. Function A is y = 4x + 2. Function B has the table: x = 0, 1, 2 and y = 5, 7, 9. Which statement is true?

    A. Function B grows faster because it starts at 5

    B. They grow at the same rate

    C. Function A grows faster: its rate of change is 4, compared with 2

    D. Function B grows faster because 9 is bigger than 4

12. Salem has AED 50 and saves AED 20 a week. Hamdan has AED 10 and saves AED 30 a week. After how many weeks do they have the same amount, and how much is it?

Section 2: Shapes, measuring and space

Standards practised: 8.G.A.1, 8.G.A.2, 8.G.A.3, 8.G.A.4, 8.G.A.5, 8.G.B.6, 8.G.B.7, 8.G.B.8, 8.G.C.9

A worked example

Find the distance between (-2, 1) and (4, 9).

1. Horizontal change: 4 - (-2) = 6. Vertical change: 9 - 1 = 8.
2. These are the two legs of a right triangle: 6 squared + 8 squared = 36 + 64 = 100.
3. The distance is the square root of 100, which is 10.

10 units

Questions

1. Translate the point (2, 3) by 4 units right and 5 units down. Where does it land?

   A. (-2, 8)

   B. (6, -2)

   C. (6, 8)

   D. (-2, -2)

2. Reflect the point (3, -4) across the x-axis, then reflect the original point across the y-axis.

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3. What is the image of (2, 5) after a rotation of 90 degrees anticlockwise about the origin?

   A. (5, -2)

   B. (-2, -5)

   C. (-5, 2)

   D. (5, 2)

4. A triangle is dilated from the origin with scale factor 3. The point (2, 4) is on it and one side is 2 cm long. Where does the point go and how long is the side now? Is the image congruent or similar?

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5. Triangle ABC has angles 40 and 60 degrees. Triangle DEF has angles 60 and 80 degrees. Are the triangles similar?

   A. Yes, both have angles 40, 60 and 80 degrees

   B. No, because only the 60 degree angles match

   C. No, because ABC has a 40 degree angle and DEF does not

   D. Yes, because both have a 60 degree angle

6. A triangular flag has two angles of 62 degrees and 47 degrees. Find the third angle.

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7. Two parallel lines are cut by a transversal. One angle is 65 degrees. Find its alternate interior angle and the same-side interior angle.

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8. A ladder leans against a wall. Its foot is 3 m from the wall and it reaches 4 m up the wall. How long is the ladder?

   A. 7 m

   B. 25 m

   C. 12 m

   D. 5 m

9. A triangle has sides 8 cm, 15 cm and 17 cm. Is it a right triangle?

   A. No, because 8 + 15 is 23, not 17

   B. Yes, because 64 + 225 = 289 = 17 squared

   C. Yes, because 8 + 15 + 17 = 40

   D. No, because the sides are all different

10. On a map grid in km, a hotel is at (2, 1) and the souq is at (7, 13). How far apart are they in a straight line?

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11. A cylindrical water tank has radius 2 m and height 5 m. Find its volume in terms of pi, then use 3.14 for pi.

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12. A cone has radius 3 cm and height 4 cm. A sphere has radius 3 cm. Find each volume (use 3.14 for pi).

Section 3: Data, graphs and chance

Standards practised: 8.SP.A.1, 8.SP.A.2, 8.SP.A.3, 8.SP.A.4

A worked example

In Grade 6, 20 students were asked about lunch: 15 bring a lunch box and 5 buy lunch. In Grade 8, 30 students were asked: 15 bring a lunch box and 15 buy lunch. Which grade is more likely to bring a lunch box?

1. The groups are different sizes, so compare percentages, not counts.
2. Grade 6: 15 out of 20 = 75%.
3. Grade 8: 15 out of 30 = 50%.
4. 75% is more than 50%.

Grade 6 (75% compared with 50%)

Questions

1. A scatter plot shows that on hotter days a shop sells more cold drinks. Describe the association.

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2. A scatter plot of the age of cars and their selling price slopes downward. What kind of association is this?

   A. Positive association

   B. Negative association

   C. No association

   D. A cluster

3. The points on a scatter plot are spread out with no pattern. What can you say?

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4. Points: (1, 2), (2, 4), (3, 6), (4, 8), (5, 20). Which point is an outlier?

   A. (4, 8)

   B. (1, 2)

   C. (5, 20)

   D. (3, 6)

5. A line of best fit for the walking time y (minutes) over x km is y = 12x + 3. Predict the time for 5 km.

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6. In y = 12x + 3, what does the slope 12 mean in this situation?

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7. The line of best fit for the cost y (AED) of a school trip for x children passes through (0, 4) and (10, 24). Write its equation.

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8. A survey of 50 students: Grade 7: 18 have a library card and 12 do not. Grade 8: 14 have a card and 6 do not. How many students have a card in all?

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9. In the same survey, what percentage of Grade 8 students have a card?

   A. 70%

   B. 30%

   C. 14%

   D. 6%

10. What percentage of Grade 7 students have a card? Compare with Grade 8.

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11. Why is it better to compare percentages than counts in that survey?

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12. Pool visits and ice cream sales both rise in hot weather. What does a strong association tell us?

    A. Ice cream makes people want to swim

    B. Swimming makes people want ice cream

    C. Both can rise because of the heat; association does not prove cause

    D. The two must be unrelated

Section 4: Numbers and calculating

Standards practised: 8.NS.A.1, 8.NS.A.2, 8.EE.A.1, 8.EE.A.2, 8.EE.A.3, 8.EE.A.4

A worked example

Write 0.666... (6 repeating) as a fraction. Then simplify 2^5 × 2^-2.

1. Let x = 0.666... Then 10x = 6.666...
2. Subtract: 10x - x = 6, so 9x = 6 and x = 6/9 = 2/3.
3. Same base, so add the exponents: 2^5 × 2^-2 = 2^3 = 8.

2/3 and 8

Questions

1. Which decimal is equal to 5/8?

   A. 0.58

   B. 0.625

   C. 1.6

   D. 0.85

2. Write 0.272727... (27 repeating) as a fraction.

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3. Which of these numbers is irrational?

   A. The square root of 16

   B. 0.5

   C. The square root of 20

   D. 1/3

4. Between which two whole numbers does the square root of 50 lie?

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5. A square garden has an area of 50 square metres. Estimate the length of one side to the nearest tenth of a metre.

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6. Which is bigger: the square root of 10 or 3.2?

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7. Simplify and evaluate: 3^4 × 3^2

   A. 3^6 = 729

   B. 3^8 = 6,561

   C. 9^6 = 531,441

   D. 3^2 = 9

8. Noura has 5^7 stickers and packs them in bundles of 5^4. How many bundles does she make?

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9. Evaluate: (2^3)^2

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10. Evaluate: 2^-3 and 10^0

    A. -8 and 0

    B. -6 and 1

    C. 1/8 and 0

    D. 1/8 and 1

11. The Hope Probe travelled about 493,000,000 km to reach Mars. Which is this number in scientific notation?

    A. 4.93 × 10^8

    B. 4.93 × 10^9

    C. 49.3 × 10^7

    D. 0.493 × 10^9

12. A solar park makes 8 × 10^6 units of energy a day and a rooftop array makes 4 × 10^3. How many times as much does the solar park make? Write the answer in scientific notation.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. Take 7 from both sides: 3x = 21. Divide by 3: x = 7. Check: 3 × 7 + 7 = 28.
2. A. x = 8. Expand the bracket: 5x - 10 = 3x + 6, so 2x = 16 and x = 8. If your child chose C (x = 16), they stopped at 2x = 16 and forgot to divide by 2.
3. Undo the adding of 3: x/2 = 11 - 3 = 8. Undo the halving: x = 8 × 2 = 16. Check: 16/2 + 3 = 11.
4. D. It has no solution. Taking 2x from both sides leaves 3 = 5, which is never true, so there is no solution. If your child chose B (x = 0), they tried one number, but every x leaves 3 = 5.
5. C. 3. Slope is rise over run: (11 - 2) / (4 - 1) = 9/3 = 3. If your child chose A (9), they found the rise but forgot to divide by the run.
6. The slope is 2 and the line crosses the y-axis at -3, so y = mx + b gives y = 2x - 3.
7. Cost is the start fee plus the rate times the km, so y = 3x + 12. For 10 km: 3 × 10 + 12 = 30 + 12 = AED 42.
8. D. (3, 7) Set the two y values equal: 2x + 1 = x + 4, so x = 3 and y = 3 + 4 = 7. If your child chose A (7, 3), they swapped x and y.
9. Add the two equations: 2x = 14, so x = 7 for Maryam. Then y = 10 - 7 = 3 for Noura. Check: 7 - 3 = 4.
10. B. No, because the input 2 has two outputs, 5 and 7. A function gives each input exactly one output, but the input 2 has two outputs, 5 and 7. If your child chose C (yes, three pairs), they counted the pairs instead of checking each input.
11. C. Function A grows faster: its rate of change is 4, compared with 2. A changes by 4 for each step in x, and B goes 5, 7, 9 so it changes by 2. If your child chose D (9 is bigger than 4), they compared single values instead of the rate of change.
12. Same amount means 50 + 20w = 10 + 30w, so 40 = 10w and w = 4. Then 50 + 20 × 4 = AED 130 each.

Section 2: Shapes, measuring and space

1. B. (6, -2) Right adds 4 to x: 2 + 4 = 6. Down takes 5 from y: 3 - 5 = -2. If your child chose C (6, 8), they added 5 to y instead of taking it away.
2. Across the x-axis only y changes sign, so (3, -4) becomes (3, 4). Across the y-axis only x changes sign, so (3, -4) becomes (-3, -4).
3. C. (-5, 2) A 90 degree anticlockwise turn sends (x, y) to (-y, x), so (2, 5) goes to (-5, 2). If your child chose A (5, -2), they turned clockwise instead.
4. Multiply each coordinate by 3: (2, 4) becomes (6, 12), and the side is 2 × 3 = 6 cm. The shape is the same but bigger, so it is similar, not congruent.
5. A. Yes, both have angles 40, 60 and 80 degrees. ABC's third angle is 180 - 40 - 60 = 80 and DEF's is 180 - 60 - 80 = 40, so both have 40, 60 and 80. If your child chose B (only the 60 degree angles match), they forgot to find the third angles.
6. The three angles add to 180: 62 + 47 = 109, and 180 - 109 = 71 degrees.
7. Alternate interior angles are equal, so it is 65 degrees. Same-side interior angles add to 180, so it is 180 - 65 = 115 degrees.
8. D. 5 m. 3 squared + 4 squared = 9 + 16 = 25, and the square root of 25 is 5 m. If your child chose A (7 m), they added 3 + 4 instead of squaring first.

9. B. Yes, because 64 + 225 = 289 = 17 squared. 8 squared + 15 squared = 64 + 225 = 289, which equals 17 squared, so it is a right triangle. If your child chose A (no, 8 + 15 is 23), they added the sides instead of squaring them.
10. The sideways change is 7 - 2 = 5 and the up change is 13 - 1 = 12. Then 25 + 144 = 169, and the square root of 169 is 13 km.
11. Volume = pi x r squared x h = pi x 4 × 5 = 20 pi cubic m. With 3.14: 20 × 3.14 = 62.8 cubic m.
12. Cone: 1/3 x pi x 9 × 4 = 12 pi, about 37.7 cubic cm. Sphere: 4/3 x pi x 27 = 36 pi, about 113 cubic cm.

Section 3: Data, graphs and chance

1. More cold drinks sell as the days get hotter, so both values rise together. That is a positive association.
2. B. Negative association. The points slope down: as age goes up, price goes down, so the association is negative. If your child chose A (positive), they matched a rising age with a rising price.
3. The points follow no line or pattern, so there is no association between the two variables.
4. C. (5, 20) The first four points follow y = 2x, so (5, 20) should be (5, 10). It is far from the pattern, so it is the outlier. If your child chose A (4, 8), they picked a point that fits y = 2x.
5. Put x = 5 into the line: 12 × 5 + 3 = 60 + 3 = 63 minutes.
6. The slope is the change in y for each 1 more in x, so each extra km adds about 12 minutes of walking.
7. Slope = (24 - 4) / (10 - 0) = 2, and the line crosses the y-axis at 4, so y = 2x + 4.
8. Add the cards of both grades: 18 + 14 = 32 students have a card.
9. A. 70%. Grade 8 has 14 + 6 = 20 students, and 14 / 20 = 70%. If your child chose C (14%), they used the count as the percentage.
10. Grade 7 has 18 + 12 = 30 students, so 18 / 30 = 60%. Grade 8 is 14 / 20 = 70%, so a greater share of Grade 8 have a card.
11. Grade 7 has 18 + 12 = 30 students and Grade 8 has 14 + 6 = 20, so the counts are not out of the same total. Percentages make the comparison fair.
12. C. Both can rise because of the heat; association does not prove cause. The heat explains both, so a strong association does not prove that one thing causes the other. If your child chose A (ice cream makes people want to swim), they read the link as cause and effect.

Section 4: Numbers and calculating

1. B. 0.625. Divide the top by the bottom: 5 / 8 = 0.625. If your child chose A (0.58), they wrote the 5 and the 8 side by side after the point.
2. Let x = 0.2727..., so 100x = 27.2727.... Subtract: 99x = 27, so x = 27/99 = 3/11.
3. C. The square root of 20. 20 is not a perfect square, so its square root never ends or repeats, which makes it irrational. If your child chose D (1/3), they thought any endless decimal is irrational, but 1/3 repeats.
4. 7 × 7 = 49 and 8 × 8 = 64, and 50 lies between 49 and 64, so the square root of 50 is between 7 and 8.
5. The side is the square root of 50, close to 7. Try 7.1: 7.1 × 7.1 = 50.41, so the side is about 7.1 m.
6. 3.2 × 3.2 = 10.24, which is more than 10, so 3.2 is the bigger number.

7. A. 3^6 = 729. Same base, so add the exponents: 3^(4 + 2) = 3^6 = 729. If your child chose B (3^8), they multiplied the exponents instead of adding.
8. Same base, so subtract the exponents: 5^7 / 5^4 = 5^(7 - 4) = 5^3 = 125 bundles.
9. Power of a power: multiply the exponents, so (2^3)^2 = 2^(3 × 2) = 2^6 = 64.
10. D. 1/8 and 1. 2^-3 = 1/2^3 = 1/8, and any non-zero number to the power 0 is 1. If your child chose B (-6 and 1), they multiplied 2 by -3 instead of taking the reciprocal.
11. A. 4.93 × 10^8. Move the decimal point 8 places left to get 4.93, so 493,000,000 = 4.93 × 10^8. If your child chose B (4.93 × 10^9), they counted one place too many.
12. Divide the numbers: 8 / 4 = 2. Subtract the powers: 10^6 / 10^3 = 10^3. So the answer is 2 × 10^3, which is 2,000 times.